

Module -3

Sinusoidal Steady State Analysis

Dr. Sathya P
Associate Professor
SENSE

Sinusoids

- A sinusoid is a signal that has the form of the sine or cosine function.
- A general expression for the sinusoid

$$v(t) = V_m \sin \omega t$$

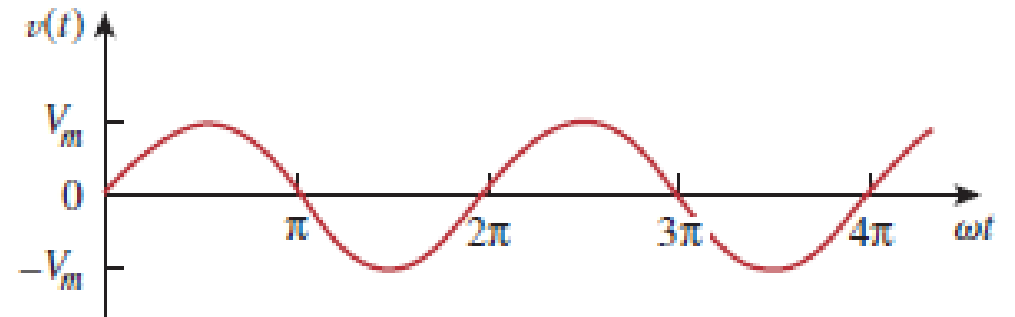
where

V_m = the *amplitude* of the sinusoid

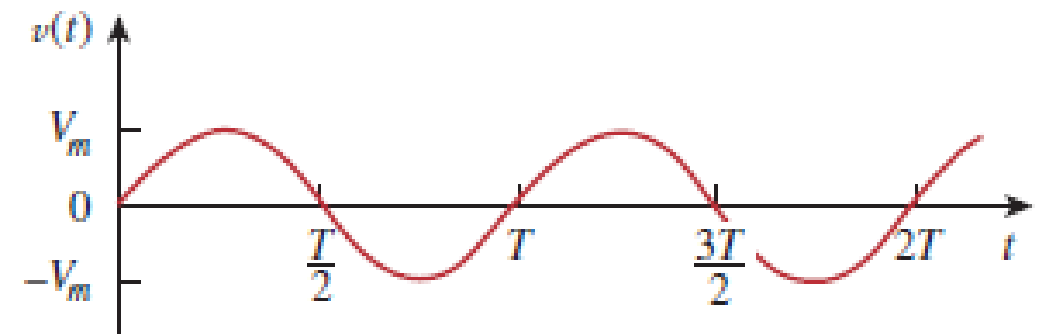
ω = the *angular frequency* in radians/s

ωt = the *argument* of the sinusoid

$$T = \frac{2\pi}{\omega}$$



A sketch of $V_m \sin \omega t$: (a) as a function of ωt



(b) as a function of t .

A **periodic function** is one that satisfies $f(t) = f(t + nT)$, for all t and for all integers n .

$$\begin{aligned}v(t + T) &= V_m \sin \omega(t + T) = V_m \sin \omega \left(t + \frac{2\pi}{\omega} \right) \\&= V_m \sin(\omega t + 2\pi) = V_m \sin \omega t = v(t)\end{aligned}$$

$$v(t + T) = v(t)$$

$$f = \frac{1}{T}$$

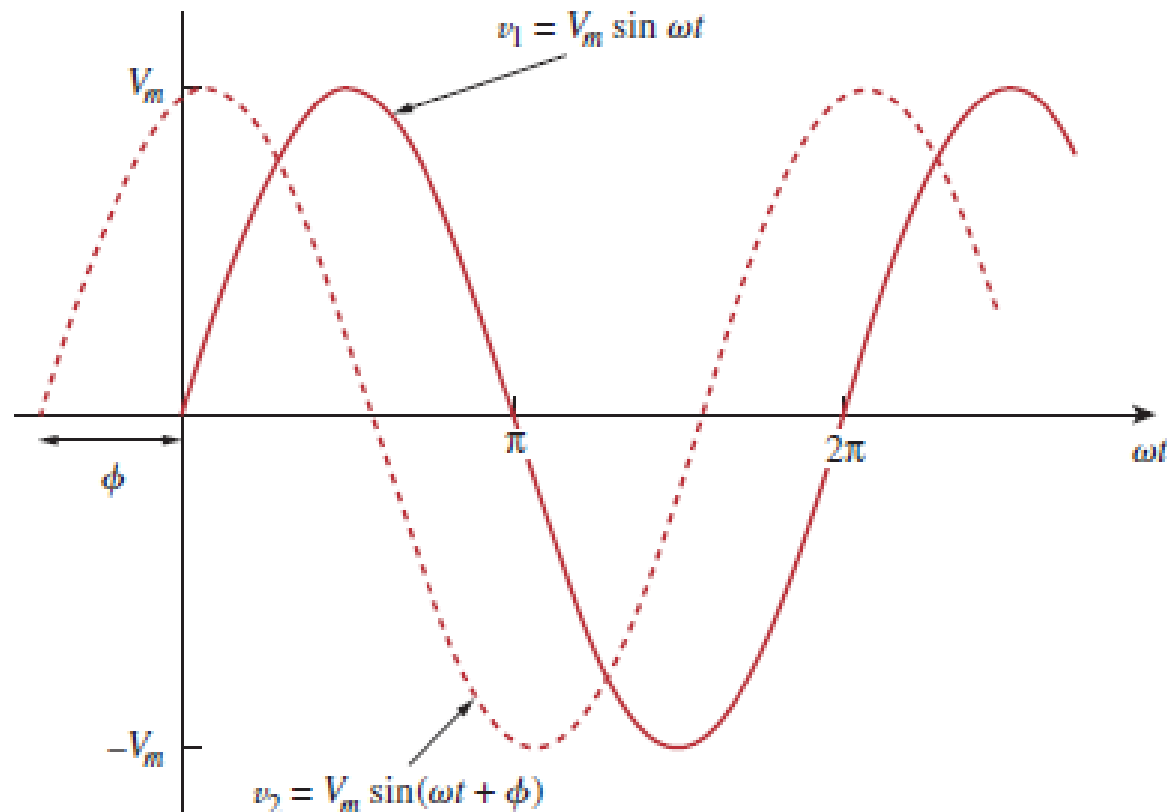
$$\omega = 2\pi f$$

While ω is in radians per second (rad/s), f is in hertz (Hz).

$$v(t) = V_m \sin(\omega t + \phi)$$

where $(\omega t + \phi)$ is the argument and ϕ is the *phase*. Both argument and phase can be in radians or degrees.

$$v_1(t) = V_m \sin \omega t \quad \text{and} \quad v_2(t) = V_m \sin(\omega t + \phi)$$



- Only two sinusoidal values with the same frequency can be compared by their amplitude and phase difference.
- If phase difference is zero, they are in phase; if phase difference is not zero, they are out of phase.

Trigonometric Identities

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

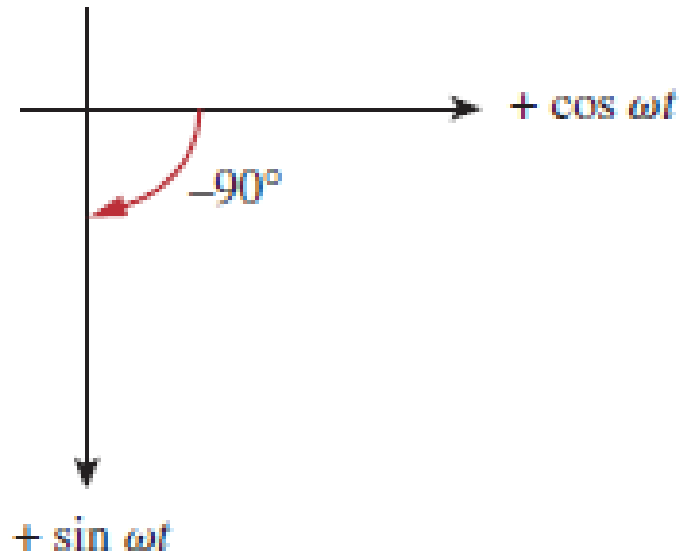
$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

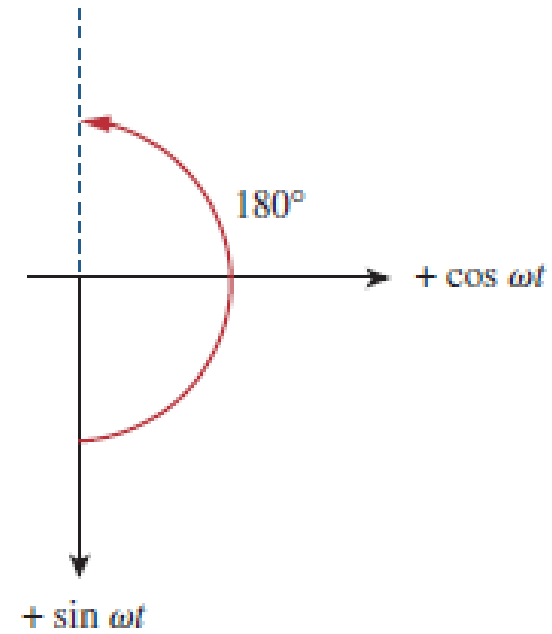
$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

Graphical Representation of a Sinusoid



(a) $\cos(\omega t - 90^\circ) = \sin \omega t$.



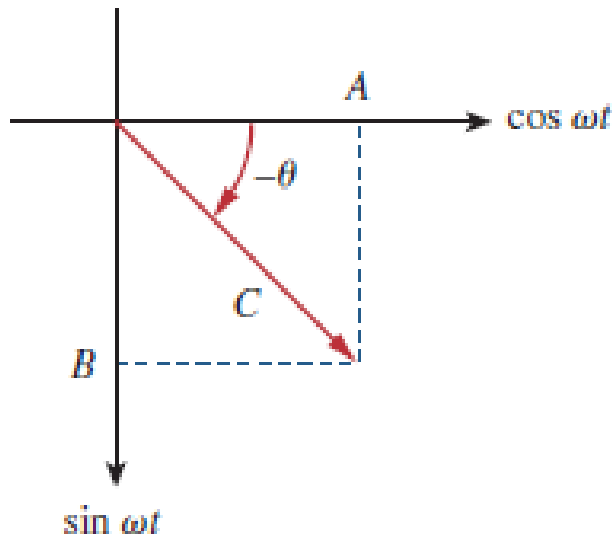
(b) $\sin(\omega t + 180^\circ) = -\sin \omega t$.

Addition of two Sinusoids

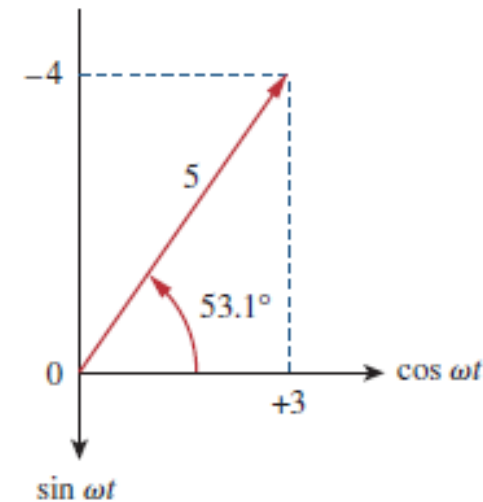
$$A \cos \omega t + B \sin \omega t = C \cos(\omega t - \theta)$$

where

$$C = \sqrt{A^2 + B^2}, \quad \theta = \tan^{-1} \frac{B}{A}$$



(a) Adding $A \cos \omega t$ and $B \sin \omega t$.



(b) adding $3 \cos \omega t$ and $-4 \sin \omega t$.

Phasors

- A phasor is a complex number that represents the amplitude and phase of a sinusoid.
- It can be represented in one of the following three forms:

$$z = x + jy \quad \text{Rectangular form}$$

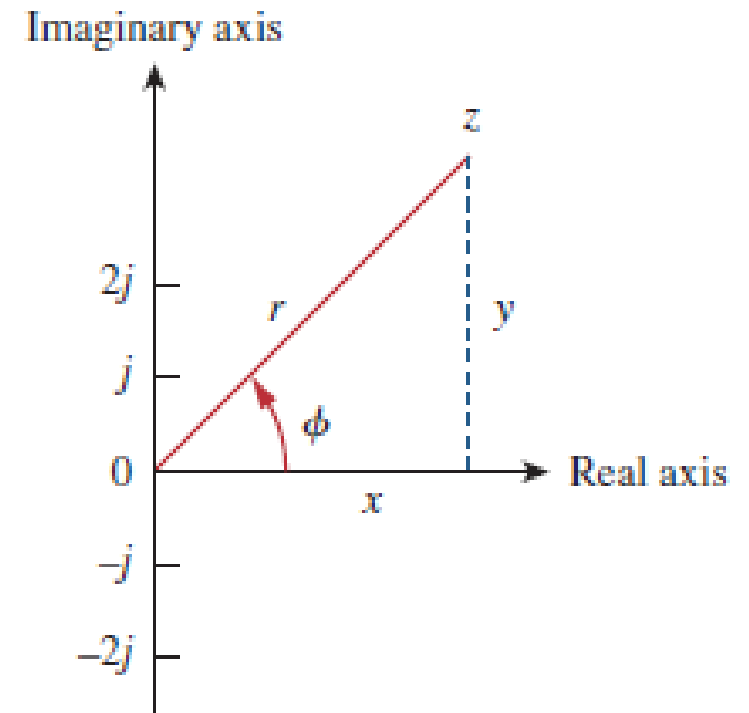
$$z = r \angle \phi \quad \text{Polar form}$$

$$z = re^{j\phi} \quad \text{Exponential form}$$

$$r = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x}$$

$$x = r \cos \phi, \quad y = r \sin \phi$$

$$z = x + jy = r \angle \phi = r(\cos \phi + j \sin \phi)$$



Mathematical Operations

Addition:

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Subtraction:

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

Multiplication:

$$z_1 z_2 = r_1 r_2 \angle \underline{\phi_1 + \phi_2}$$

Division:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle \underline{\phi_1 - \phi_2}$$

Reciprocal:

$$\frac{1}{z} = \frac{1}{r} \angle \underline{-\phi}$$

Square Root:

$$\sqrt{z} = \sqrt{r} \angle \underline{\phi/2}$$

Complex Conjugate:

$$z^* = x - jy = r \angle \underline{-\phi} = re^{-j\phi}$$

$$\frac{1}{j} = -j$$

Phasor Domain Representation

$$\begin{array}{ccc} v(t) = V_m \cos(\omega t + \phi) & \Leftrightarrow & \mathbf{V} = V_m \angle \phi \\ \text{(Time-domain} & & \text{(Phasor-domain} \\ \text{representation)} & & \text{representation)} \end{array}$$

Sinusoid-phasor transformation.

Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90^\circ$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90^\circ$

$$\begin{array}{ccc} \frac{dv}{dt} & \Leftrightarrow & j\omega V \\ \text{(Time domain)} & & \text{(Phasor domain)} \end{array}$$

$$\begin{array}{ccc} \int v \, dt & \Leftrightarrow & \frac{V}{j\omega} \\ \text{(Time domain)} & & \text{(Phasor domain)} \end{array}$$

The differences between $v(t)$ and V should be emphasized:

1. $v(t)$ is the *instantaneous or time domain* representation, while V is the *frequency or phasor domain* representation.
2. $v(t)$ is time dependent, while V is not.
3. $v(t)$ is always real with no complex term, while V is generally complex.

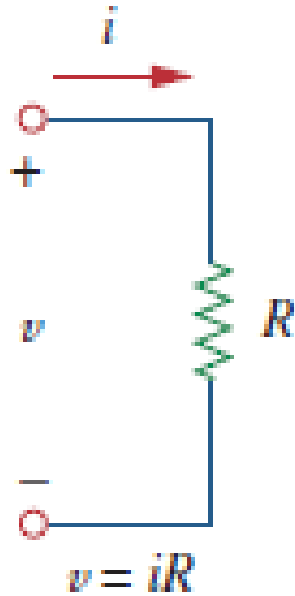
Phasor Relationships for Circuit Elements

- Resistor:

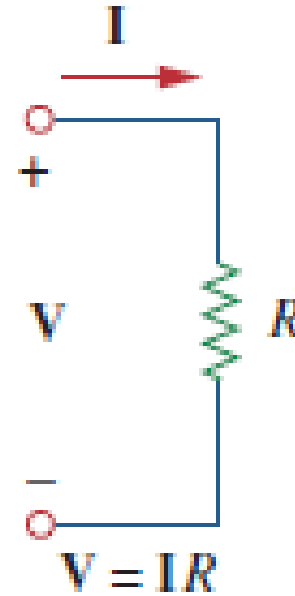
$$v = iR = RI_m \cos(\omega t + \phi)$$

$$\mathbf{V} = RI_m \angle \phi$$

$$\mathbf{V} = R\mathbf{I}$$

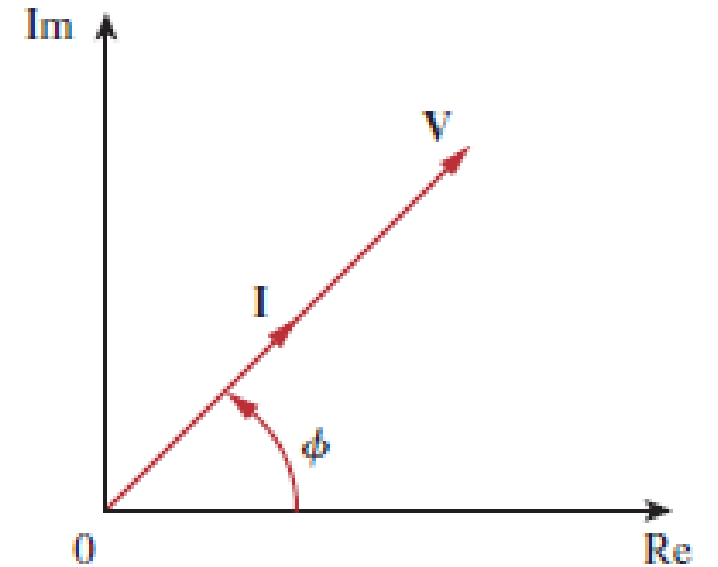


(a)



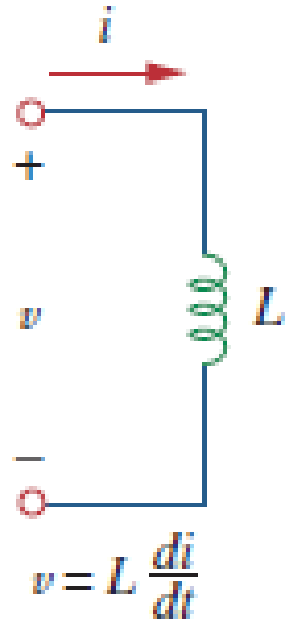
(b)

Voltage-current relations for a resistor in the: (a) time domain, (b) frequency domain.

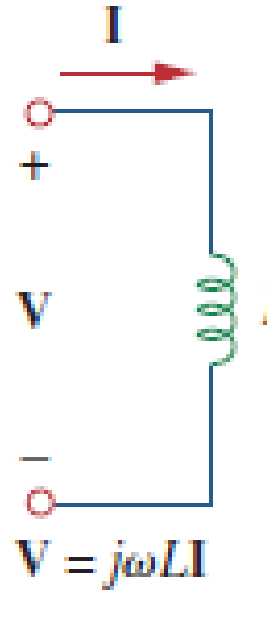


Phasor diagram for the resistor.

- Inductor:

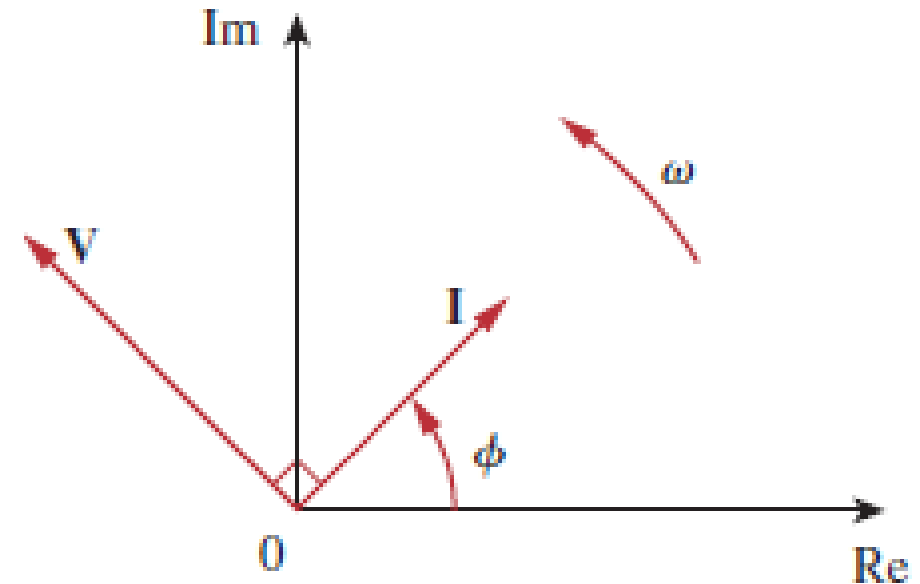


(a)



(b)

Voltage-current relations for an inductor in the: (a) time domain, (b) frequency domain.

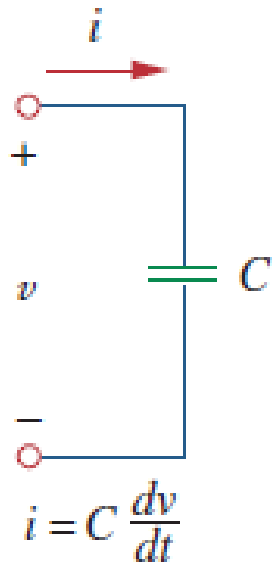


Phasor diagram for the inductor; I lags V .

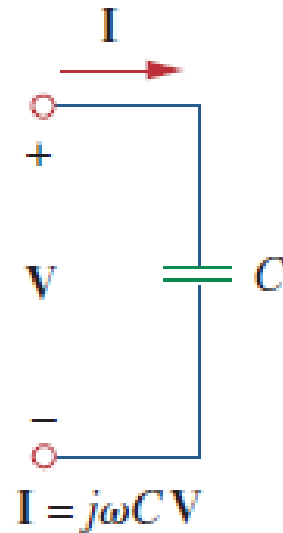
- Capacitor:

$$i = C \frac{dv}{dt}$$

$$\mathbf{I} = j\omega C \mathbf{V} \quad \Rightarrow \quad \mathbf{V} = \frac{\mathbf{I}}{j\omega C}$$

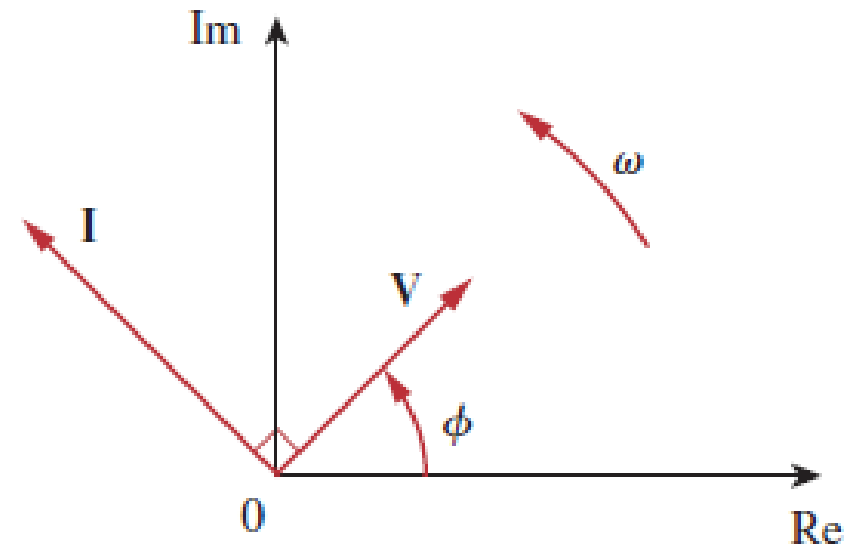


(a)



(b)

Voltage-current relations for a capacitor in the: (a) time domain, (b) frequency domain.



Phasor diagram for the capacitor; $\mathbf{I}$ leads $\mathbf{V}$.

Summary of voltage-current relationships.

Element	Time domain	Frequency domain
R	$v = Ri$	$\mathbf{V} = R\mathbf{I}$
L	$v = L\frac{di}{dt}$	$\mathbf{V} = j\omega L\mathbf{I}$
C	$i = C\frac{dv}{dt}$	$\mathbf{V} = \frac{\mathbf{I}}{j\omega C}$

Impedance

The **impedance** $\mathbf{Z}$ of a circuit is the ratio of the phasor voltage $\mathbf{V}$ to the phasor current $\mathbf{I}$, measured in ohms (Ω).

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}}$$

Rectangular form:

$$\mathbf{Z} = R + jX$$

where $R = \text{Re } \mathbf{Z}$ is the *resistance* and $X = \text{Im } \mathbf{Z}$ is the *reactance*.

Polar form:

$$\mathbf{Z} = |\mathbf{Z}| \angle \theta$$

$$\mathbf{Z} = R + jX = |\mathbf{Z}| \angle \theta$$

$$|\mathbf{Z}| = \sqrt{R^2 + X^2}, \quad \theta = \tan^{-1} \frac{X}{R}$$

$$R = |\mathbf{Z}| \cos \theta, \quad X = |\mathbf{Z}| \sin \theta$$

Admittance

The **admittance** $\mathbf{Y}$ is the reciprocal of impedance, measured in siemens (S).

$$\mathbf{Y} = \frac{1}{\mathbf{Z}} = \frac{\mathbf{I}}{\mathbf{V}}$$

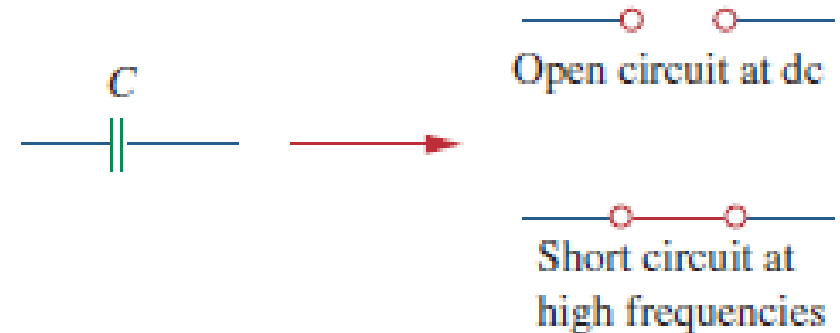
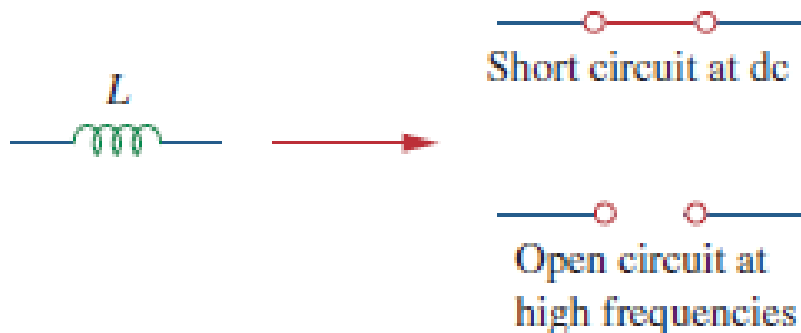
$$\mathbf{Y} = G + jB$$

where $G = \text{Re } \mathbf{Y}$ is called the *conductance*

$B = \text{Im } \mathbf{Y}$ is called the *susceptance*.

Impedances and admittances of Passive Elements

Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$



Numerical -1

Find the amplitude, phase, period, and frequency of the sinusoid

$$v(t) = 12 \cos(50t + 10^\circ)$$

Solution:

The amplitude is $V_m = 12$ V.

The phase is $\phi = 10^\circ$.

The angular frequency is $\omega = 50$ rad/s.

The period $T = \frac{2\pi}{\omega} = \frac{2\pi}{50} = 0.1257$ s.

The frequency is $f = \frac{1}{T} = 7.958$ Hz.

Numerical -2

Given the sinusoid $30 \sin(4\pi t - 75^\circ)$, calculate its amplitude, phase, angular frequency, period, and frequency.

Solution:

The amplitude is $V_m = 30\text{V}$.

The phase is $\phi = -75^\circ$.

The angular frequency is $\omega = 12.57 \text{ rad/s}$

The period $T = \frac{2\pi}{\omega} = 0.5 \text{ s}$

The frequency is $f = \frac{1}{T} = 2 \text{ Hz}$.

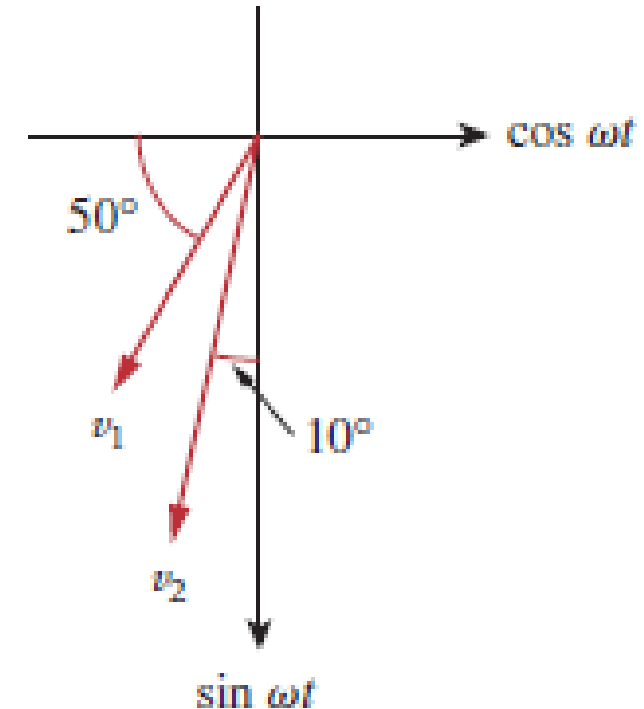
Numerical -3

Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.

Solution:

$$\begin{aligned} v_1 &= -10 \cos(\omega t + 50^\circ) = 10 \sin(\omega t + 50^\circ - 90^\circ) \\ &= 10 \sin(\omega t - 40^\circ) = 10 \sin(\omega t - 10^\circ - 30^\circ) \end{aligned}$$

But $v_2 = 12 \sin(\omega t - 10^\circ)$. Comparing the two shows that v_1 lags v_2 by 30° . This is the same as saying that v_2 leads v_1 by 30° .



Numerical -4

Evaluate the complex number: $(40\angle 50^\circ + 20\angle -30^\circ)^{1/2}$

Solution:

Using polar to rectangular transformation,

$$40\angle 50^\circ = 40(\cos 50^\circ + j \sin 50^\circ) = 25.71 + j30.64$$

$$20\angle -30^\circ = 20[\cos(-30^\circ) + j \sin(-30^\circ)] = 17.32 - j10$$

Adding them up gives

$$40\angle 50^\circ + 20\angle -30^\circ = 43.03 + j20.64 = 47.72\angle 25.63^\circ$$

Taking the square root of this,

$$(40\angle 50^\circ + 20\angle -30^\circ)^{1/2} = 6.91\angle 12.81^\circ$$

Numerical - 5

Transform these sinusoids to phasors:

(a) $i = 6 \cos(50t - 40^\circ)$ A

(b) $v = -4 \sin(30t + 50^\circ)$ V

Solution:

(a) $i = 6 \cos(50t - 40^\circ)$ has the phasor

$$\mathbf{I} = 6 \angle -40^\circ \text{ A}$$

(b) Since $-\sin A = \cos(A + 90^\circ)$,

$$\begin{aligned} v &= -4 \sin(30t + 50^\circ) = 4 \cos(30t + 50^\circ + 90^\circ) \\ &= 4 \cos(30t + 140^\circ) \text{ V} \end{aligned}$$

The phasor form of v is

$$\mathbf{V} = 4 \angle 140^\circ \text{ V}$$